

AMENDMENT NO. 1 FEBRUARY 2001
TO
IS 7401 : 1987 SPECIFICATION FOR PARAFFIN WAX
FOR EXPLOSIVES AND PYROTECHNIC INDUSTRY
(First Revision)

(*Page 1, clause 3.2*) — Substitute the following for the existing clause:

'3.2 The material shall also comply with the requirements laid down in Table 1 when tested according to the methods prescribed in P series of IS 1448*, IS 5741† and Appendix A to this standard. Reference to the relevant clauses of Appendix A, relevant P series of IS 1448* and IS 5741 : 1970† is given in columns 5, 6 and 7 of the table.'

(*Page 1, footnote*) — Insert the following footnote at the end:

†Methods for determination of pH.'

(*Page 1, Table 1*) — Substitute table given on page 2 for the existing table:

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**Table 1 Requirements for Paraffin Wax for
Explosive and Pyrotechnic Industry**

Sl No.	Characteristic	Requirement for		Method of Test, Ref to		
		Soft Grade	Hard Grade	CP No in Appendix A	(P) of IS 1448*	No.
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Melting point (cooling curve), range, °C	52-56	60-64	A-2	—	—
ii)	Acidity:					
	a) Organic (mg of KOH per g), <i>Max</i>	0.10	0.10	—	P:2	—
	b) Inorganic (mg of KOH per g), <i>Max</i>	0.02	0.02	—	P:2	—
iii)	Loss on drying at 90°C, percent by mass, <i>Max</i>	0.5	0.5	A-3	—	—
iv)	Penetration (by Vicat needle), mm, <i>Max</i>	3	1	A-4	—	—
v)	Saponification value, <i>Max</i>	1.0	1.0	—	P:55	—
vi)	Ask, percent by mass, <i>Max</i>	0.03	0.33	—	P:4	—
vii)	Oil content, percent by mass, <i>Max</i>	1.0	0.025	A-5	—	—
viii)	Matter insoluble in petroleum ether (60-90°C BP range), percent by mass, <i>Max</i>	—	0.05	A-7	—	—
ix)	pH of 5 percent of water extract in 100 ml of boiled antral	—	6.0-7.5	—	—	5741: 1970†
*Methods of test for petroleum and its products.						
†Methods for determination of pH.						

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(Page 9, clause A-6.6.1) — Insert the following new clause A-7 at the end:

A-7 DETERMINATION OF MATTER INSOLUBLE IN PETROLEUM ETHER

A-7.1 Outline of the Method — The sample is dissolved in petroleum ether (60-90°C BP range). The solution is then filtered and the residue is dried and weighed to constant mass.

A-7.2 Procedure — Weigh accurately about 5 g of the sample and digest with 100 ml of petroleum ether (60-90°C BP range) in a covered vessel on a steam bath until all the wax is dissolved. Filter through a tared, dried gooch crucible and wash it several times with warm petroleum ether to remove all the ether solubles. Dry the gooch crucible with residue and weigh to constant mass.

A-7.3 Calculation

Matter insoluble in petroleum ether,
percent by mass =

$$\frac{100\ m}{M}$$

where

m = mass, in g, of matter insoluble in petroleum ether, and

M = mass, in g, of the material taken for the test.

(CHD 26)